Amino acids and Proteins

By the end of this lecture the student will be able to :

a1. Define proteins and their functions

a2. List classifications of amino acids

a3. List the higher orders of protein structure

a4. Define denaturation and the causative agents and its effects

a5. List classification of proteins

- Proteins are polymers formed of amino acids linked together by peptide bond. The term (protein) describe molecules greater than 50 amino acids.

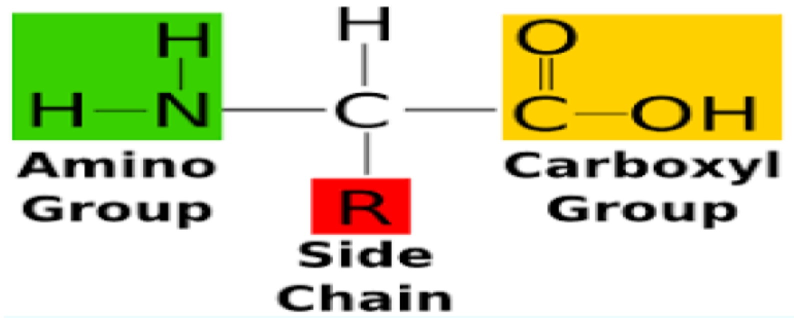
Dietary proteins

- Dietary proteins supply essential amino acids needed to synthesize structural proteins, enzymes, antibodies, some hormones and other metabolically active compounds.
- The daily requirement for proteins is 0.8 g/Kg body weight. More dietary protein is required in pregnancy, lactation, tissue repair after injury, recovery from illness and increased physical activity.

Functions of proteins

- **Nutritional:** each 1 gram protein gives about 4 Kcal.
- **Catalytic role**: All enzymes are proteins in nature.
- **Hormonal role**: Most of hormones and all cellular receptors are protein in nature.
- **Immunological role**: The antibodies (immunoglobulins) that play an important role in the body’s defensive mechanisms are proteins in nature.
- **Structural role (e.g. collagen)**
- **Plasma proteins** are responsible for most effective **osmotic pressure** of the blood

Amino Acids



- ✓ Although about 300 amino acids exist in nature, only 20 of them can polymerize in human protein structure .
- ✓ All amino acids present in mammals are L-amino acids (amino group is on the left side configuration).
- ✓ The metabolizable forms of them are L-amino acids

Classification of Amino Acids

A) Chemical classification: the number of amino groups or carboxyl groups in the amino acid:

- **Neutral amino acids** mono-amino, mono-carboxylic.
- **Acidic amino acids:** mono-amino, dicarboxylic.
- **Basic amino acids** : diamino, mono-carboxylic.

I) Neutral amino acids

- They contain one amino group and one carboxyl group. They have **5** types:

1- **Aliphatic amino acids:** e.g., Glycine, Alanine.

2- Hydroxy amino acids: contain –OH group in their side chain e.g., serine, threonine, tyrosine.

3- **Aromatic amino acids:** e.g., phenylalanine, tyrosine.

4- Sulfur-containing amino acids: e.g., Cysteine, homocysteine, cystine and methionine.

5- Heterocyclic amino acids: e.g., Tryptophan, histidine

6- Imino acids : proline and hydroxyproline

II) Acidic amino acids

They contain 2 carboxyl groups and one amino group, e.g., glutamic acid and asparatic acid.

III) Basic amino acids

They contain 2 amino groups and one carboxyl group,

e.g,.

- Arginine.
- Lysine & Hydroxylysine.
- Two amino acids not present in proteins structure are
- Citrulline.
- Ornithine.

b) Metabolic Classification: the fate of amino acid inside the body:

- **Glucogenic amino acids** that can be converted to glucose.
- **Ketogenic amino acids** that can be converted to ketone bodies.
- **Mixed amino acids**, i.e., can be converted to both glucose and ketone bodies.

Ketogenic	*& Ketogenic glucogenic*	*Glucogenic*
Leucine	**Lysine**	**Rest of amino acids**
	Isoleucine	
	Tyrosine	
	Tryptophan	
	Phenyl alanine	

Classification of Amino Acids

C) Biological classification: whether the amino acids can be synthesized in human body or not:

- **Essential amino acids:** Not synthesized in the body and must be supplied in the diet.
- **Non-essential amino acids:** Can be synthesized in the body and hence is not essential to be present in diet.

A- Essential amino acids:

- The main source for these amino acids are animal proteins (milk, egg, meat, liver, fish, chicken) and a few plant proteins (bean and lintels). They are as follows
- I Left Home To Make Visit Through London, Phlebine, Arganteen

Isoleucine	**Lysine**	**<u>Histidine</u>**	**Tryptophan**	**Methionine**
Valine	**Threonine**	**Leucine**	**Phenylalanine**	**<u>Arginine</u>**

- **B- Non essential amino acids:**

- The rest of amino acids can be synthesized inside the human body and their deficiency in diet does not affect the growth or the health.

Amino Acids

Essential

Isoleucine
Leucine
Lysine
Methionine
Phenylalanine
Threonine
Tryptophan
Valine

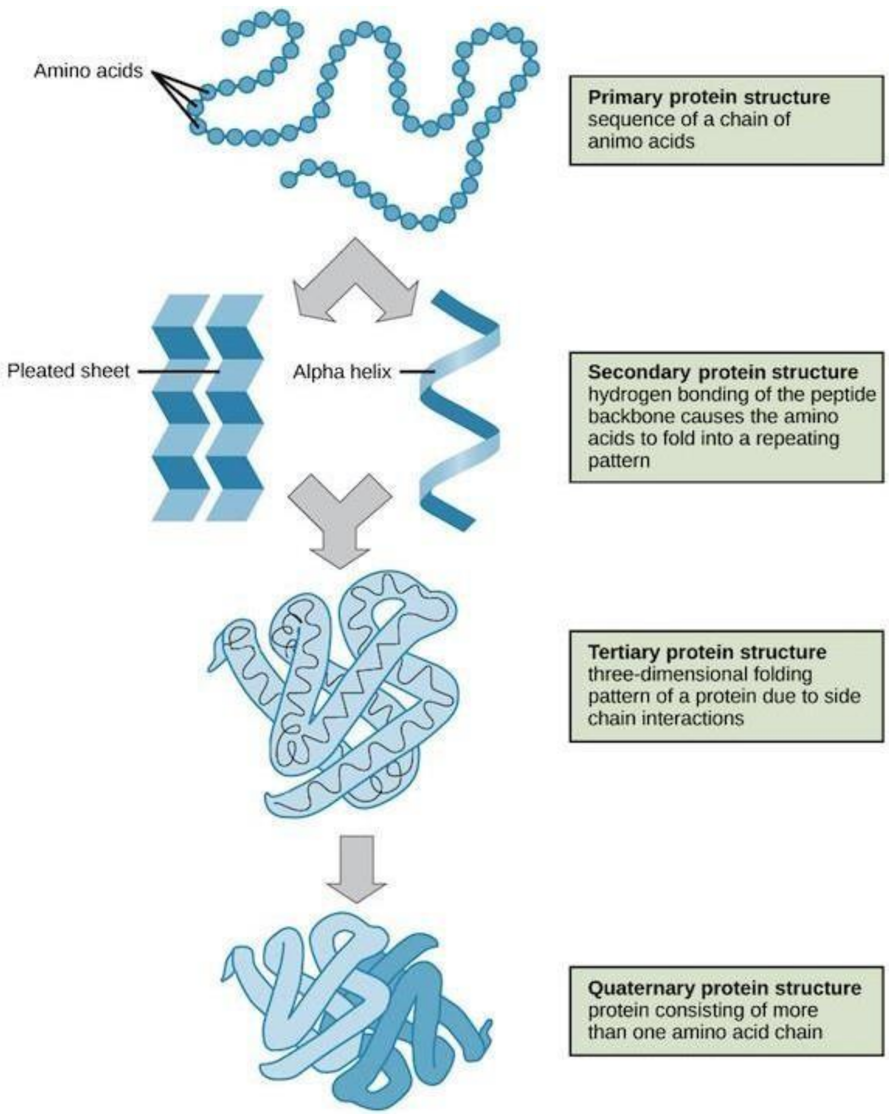
Nonessential

Alanine
Asparagine
Aspartic acid
Cysteine
Glutamic acid
Glutamine
Glycine
Proline
Serine
Tyrosine

Semi-essential

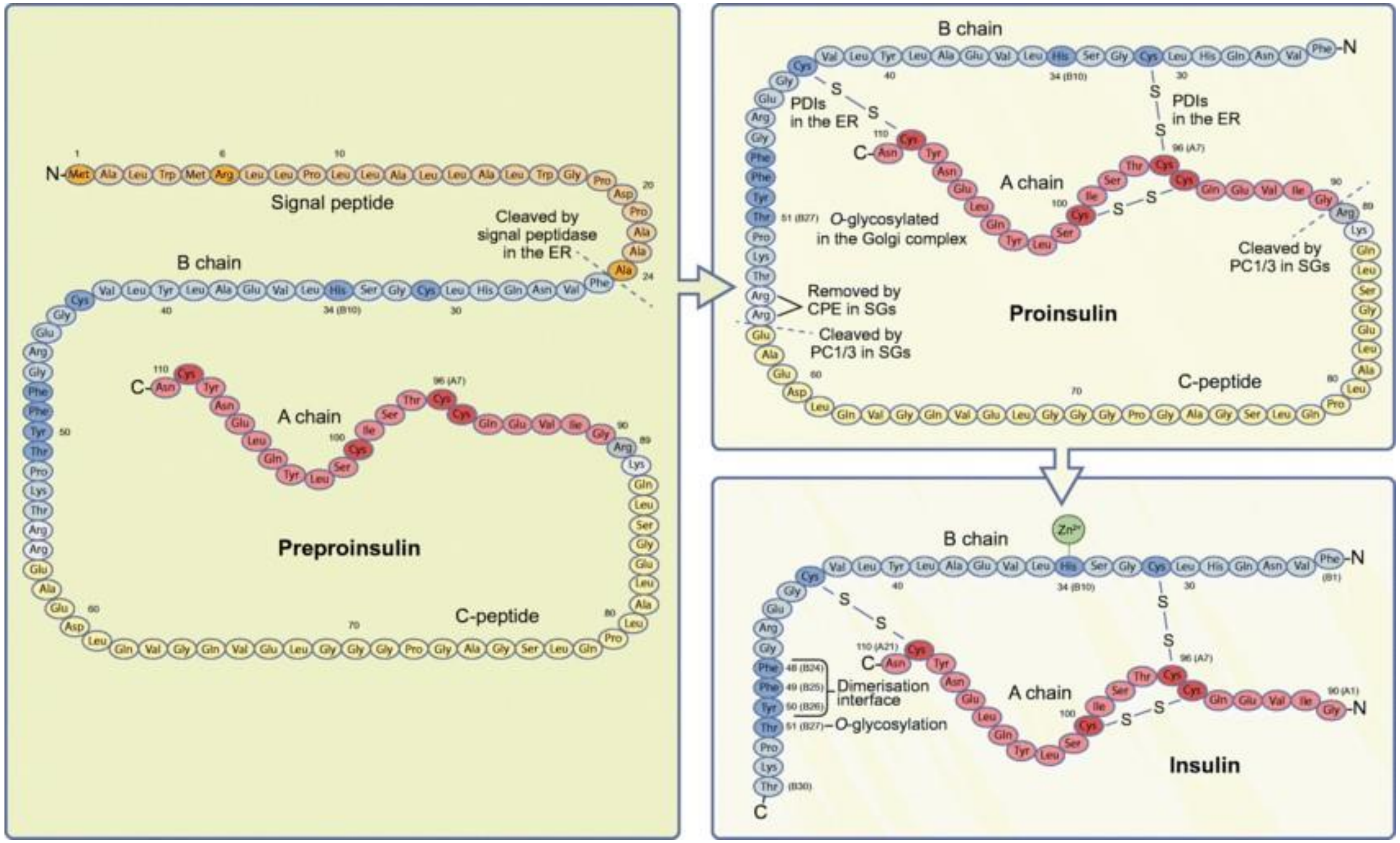
Arginine
Histidine

Structural organization of Proteins

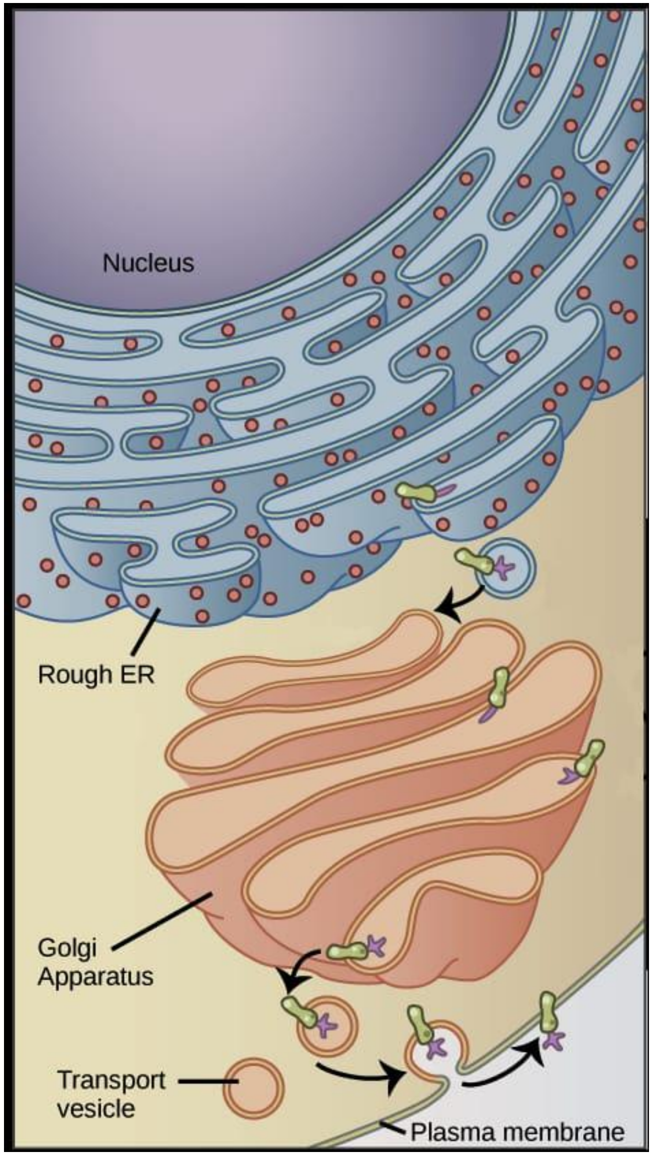


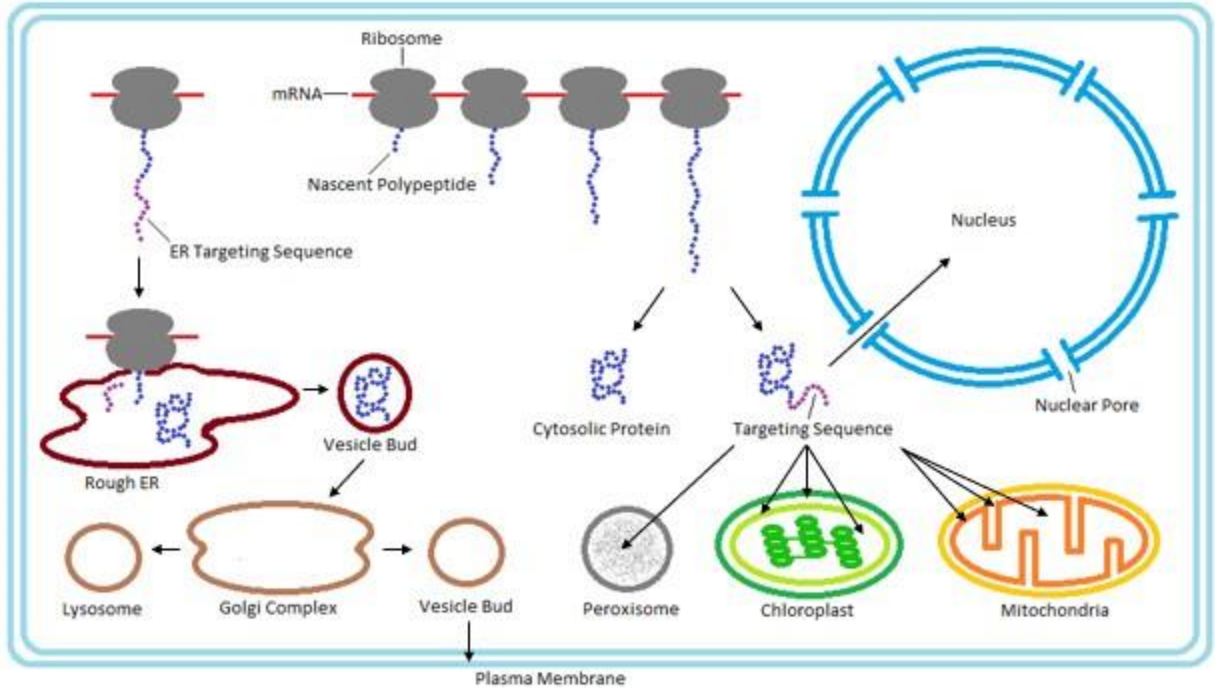
Levels of Organization

- **Primary structure**
  – Amino acid sequence of the protein
- **Secondary structure**
  – formed by H bonds in the peptide chain backbone
    - α-helix and β-sheets
- **Tertiary structure**
  – non-covalent interactions between the R groups within the protein
- **Quarternary structure**
  – Interaction between 2 polypeptide chains



The insulin peptide. Insulin is synthesised as a 110 amino-acid-long preproinsulin including a signal peptide (orange), a B chain (blue), a connecting peptide (C-peptide, yellow) and an A chain (red). The signal peptide targets the preproinsulin to the ER, where it is cleaved by the signal peptidase and converted into proinsulin. In the ER, three disulfide bonds are formed between cysteine residues with the help of PDIs. Proinsulin is trafficked from the ER, through the Golgi and the trans-Golgi network to secretory granules (SGs), where PC1/3 and CPE process the dibasic residues (grey) to form mature insulin. Zn^{2+} non-covalently binds to the HisB10 to form the insulin hexamer. Amino acids mentioned in the text are shown in a darker colour. This figure is available as part of a downloadable slideset





Protein targeting after mRNA translation

Denaturation of proteins

Definition: It is the destruction of the secondary, tertiary and quaternary structures of a protein molecule.

Causes: 1- Physical agents: - as heat, repeated thawing & freezing, high pressure, X-rays and ultraviolet rays.

2- Chemical agents: - as alcohol, strong acids and alkalies.

3-Mechanical agents: grinding

Effects of denaturation :-

1- Loss of catalytic activity of enzymes

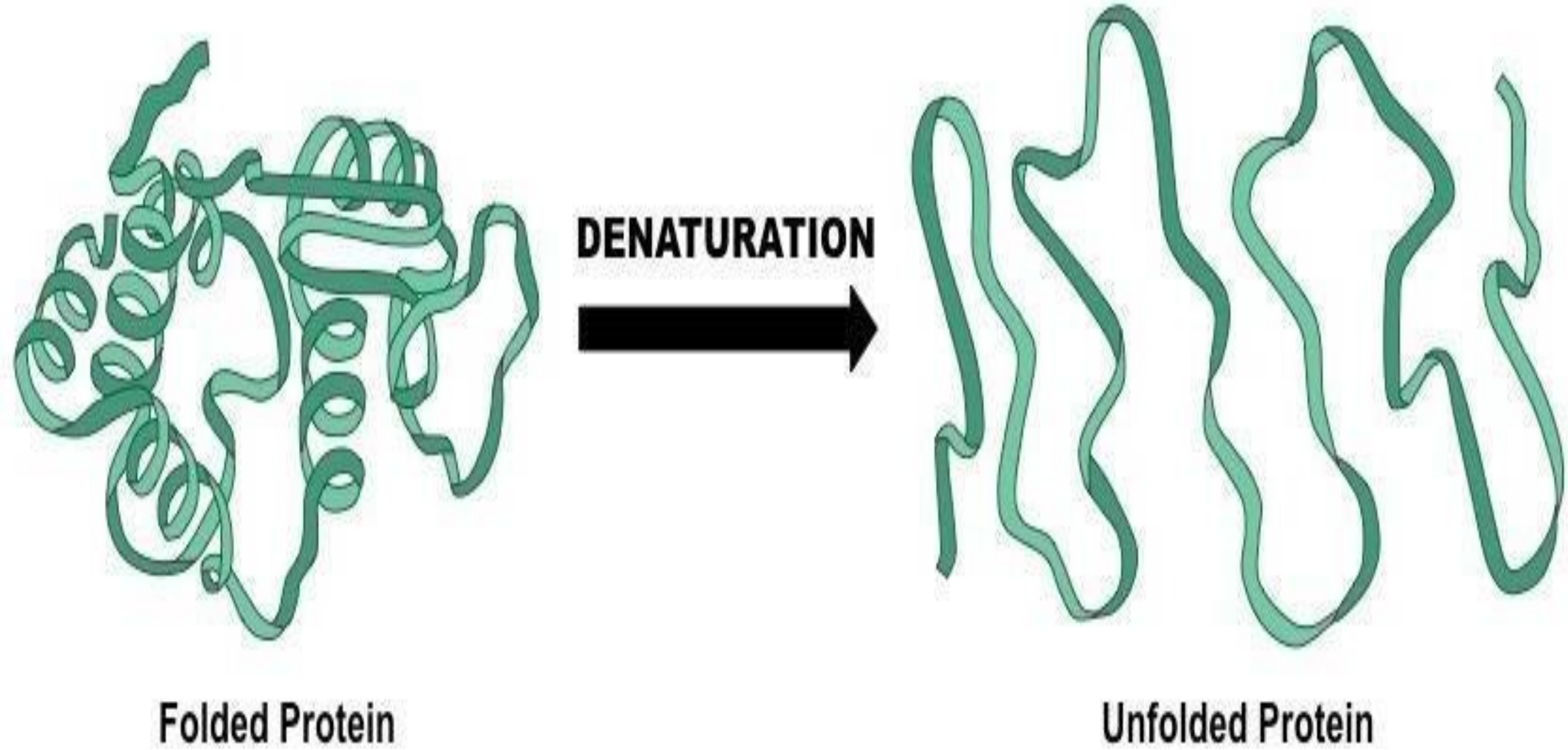
2- Loss of biological functions of hormones

3-Decreased solubility

4-Increased viscosity

5-Increased digestibility e.g. cooking meat & eggs by boiling

Denaturation of proteins



Significance & applications of Denaturation

- Denatured proteins, e.g., cooked meat are easily digested.
- Avoidance of denaturation is important for biological samples used for determination of enzymatic, hormonal or protein contents.
- Detection of albumin in urine by heat coagulation test is based on denaturation by heat.

Classification of Proteins

I. According to the biological value of the protein:

1) **Proteins of high biological value:**

   Contain all essential amino acids, and easily digested. E.g. albumin, globulins and glutelines.

2) **Proteins of low biological value**

   Lack one or more of essential amino acids

II. According to the axial ratio of the protein molecule:

1. Fibrous proteins. 2. Globular proteins.

III. According to the composition of the Protein:

A) Simple proteins. B) Conjugated proteins.

C) Derived proteins.

Two classes of proteins are classified on the basis of their axial ratio (ratio of length to breadth):

(a) Globular proteins: have axial ratio less than 10. They are compactly folded polypeptide chains like insulin, albumin and globulins.

(b) Fibrous proteins: have an axial ratio greater than 10 . The polypeptide chains coiled in spiral or helix and cross linked by hydrogen bonds like keratin, collagen and fibrin

A) Simple Proteins

These are proteins which on hydrolysis produce amino acids only.

- They include:

1. Albumins & globulins.
2. Protamines as salmin in salmon fish
3. Histones found in the nucleus
4. Gliadins "Prolamines ."found in wheat and maiz (zein).
5. Glutelins in wheat (High biological value protein)
6. Scleroproteins include collagen, elastin

Albumin & Globulins

- They are heat coagulable.
- They are of high biological value
- Examples for albumin include: ovalbumin in egg white, lactalbumin in milk and serum albumin in blood plasma.
- for globulins include : ovaglobulins in egg white, lactoglobulins in milk and plasma globulins in blood plasma.
- **Functions**

1- Both function as protein carrier for transport of different elements, vitamins & hormones.

2- Albumin keeps blood osmotic pressure.

3- Antibodies are γ-globulins.

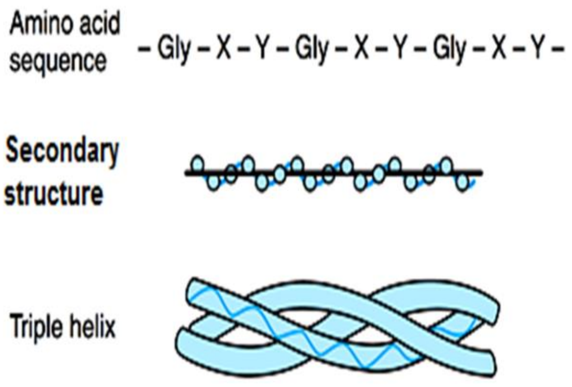
-

Scleroproteins

Collagen, is the major component of most connective tissues structure , constitutes about 25% of the protein of mammals (the most abundant one). It is found in tendons, ligaments, cartilage, bone and skin.

Structure of collagen

- ✓ All collagen types have a triple helical structure.
- ✓ It is composed of 3 chains called alpha chain.



Primary, secondary, and tertiary structures of collagen

Elastin

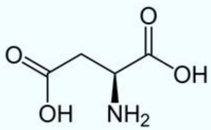
- Elastin is a connective tissue protein that is responsible for properties of extensibility and elastic recoil in tissues.
- Although not as widespread as collagen, elastin is present in large amounts in tissues that require these physical properties, e.g. lung, large arterial blood vessels, and elastic ligaments.

PEPTIDE
SYNTHESIS & ENGINEERING

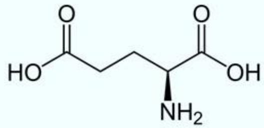
Amino acid reference chart

Download peptide calculator and AA chart poster

ACIDIC

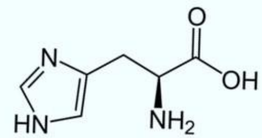


Aspartic acid (Asp, D)
MW: 115,0269
pKa: 1,99/9,90/3,90 – pI: 2,98

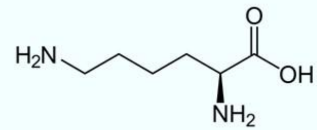


Glutamic acid (Glu, E)
MW: 129,0426
pKa: 2,10/9,47/4,07 – pI: 3,08

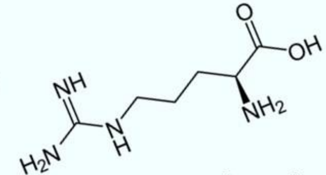
BASIC



Histidine (His, H)
MW: 137,0589
pKa: 1,80/9,33/6,04 – pI: 7,64

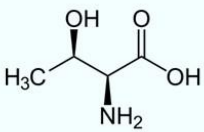


Lysine (Lys, K)
MW: 128,0950
pKa: 2,16/9,06/10,54 – pI: 9,47

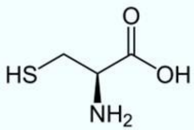


Arginine (Arg, R)
MW: 156,1011
pKa: 1,82/8,99/12,48 – pI: 10,76

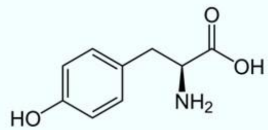
POLAR, UNCHARGED



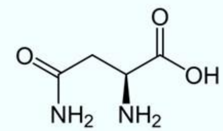
Threonine (Thr, T)
MW: 101,0477
pKa: 2.09/9.10/NA – pI: 5,60



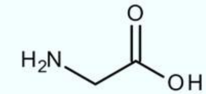
Cysteine (Cys, C)
MW: 103,0092
pKa: 1.92/10,70/8,37 – pI: 5,15



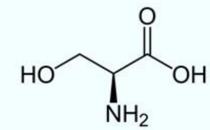
Tyrosine (Tyr, Y)
MW: 163,0633
pKa: 2,20/9,21/10,46 – pI: 5,63



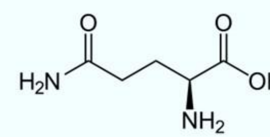
Asparagine (Asn, N)
MW: 114,0429
pKa: 2,14/8,72/NA – pI: 5,43



Glycine (Gly, G)
MW: 57,0215
pKa: 2.35/9.78/NA – pI: 6.06

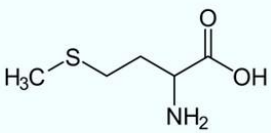


Serine (Ser, S)
MW: 87,0320
pKa: 2.19/9.21/NA – pI: 5,70

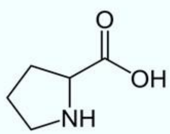


Glutamine (Gln, Q)
MW: 128,0586
pKa: 2,17/9,13/NA – pI: 5,65

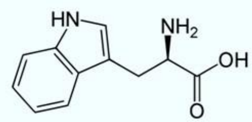
NON-POLAR, HYDROPHOBIC



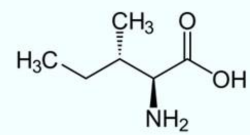
Methionine (Met, M)
MW: 131,0405
pKa: 2.13/9.28/NA – pI: 5,71



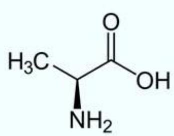
Proline (Pro, P)
MW: 97,0528
pKa: 1.95/10,64/NA – pI: 6,30



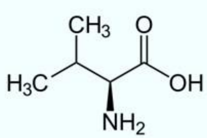
Tryptophan (Trp, W)
MW: 186,0793
pKa: 2,46/9,41/NA – pI: 5,88



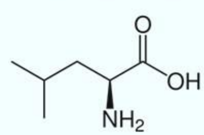
Isoleucine (Ile, I)
MW: 113,0841
pKa: 2.32/9.76/NA – pI: 6,04



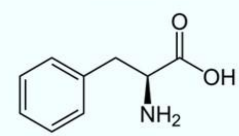
Alanine (Ala, A)
MW: 71,0371
pKa: 2.35/9.87/NA – pI: 6.11



Valine (Val, V)
MW: 99,0684
pKa: 2.29/9.74/NA – pI: 6.02

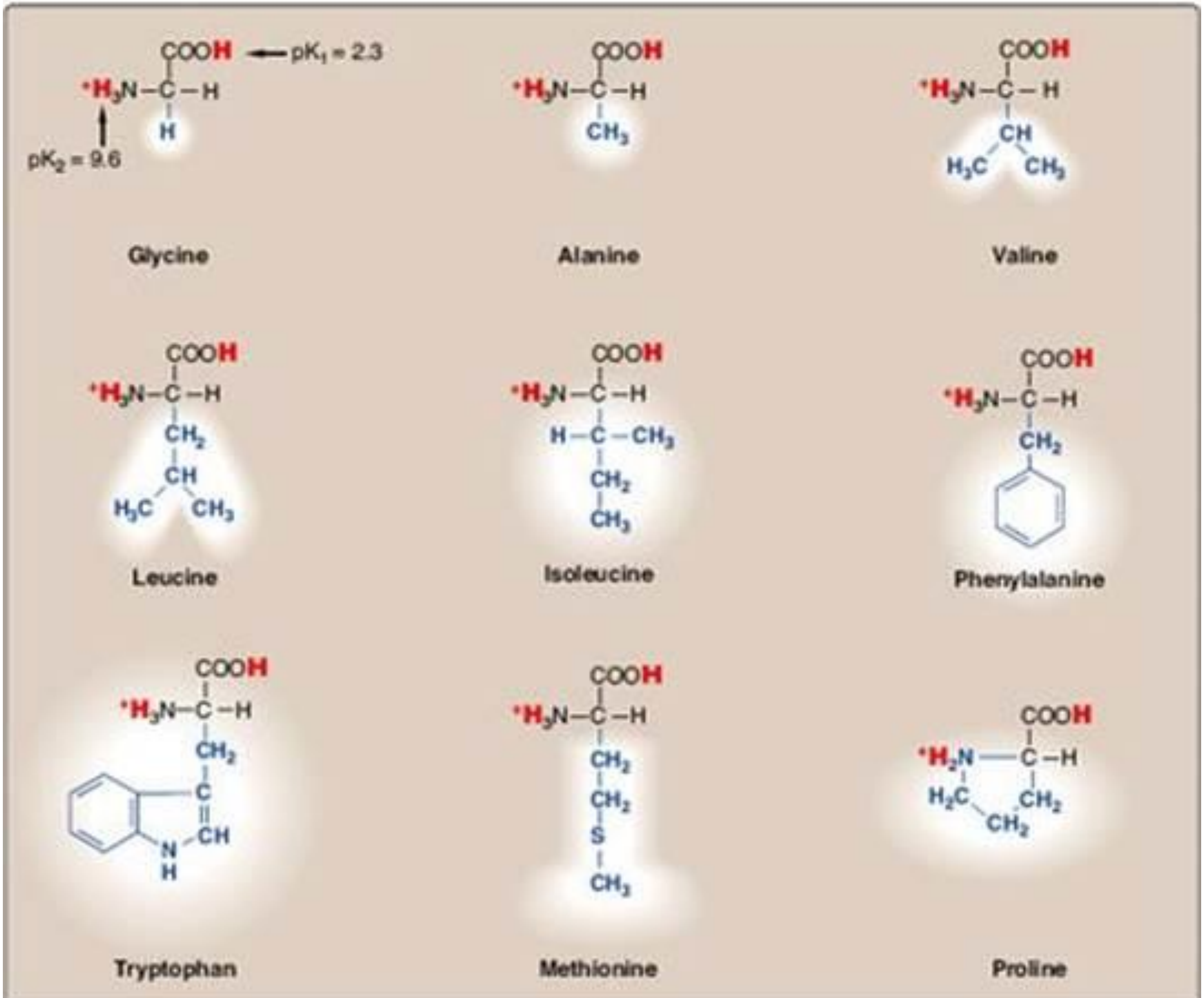


Leucine (Leu, L)
MW: 113,0841
pKa: 2.33/9.74/NA – pI: 6.04



Phenylalanine (Phe, F)
MW: 147,0684
pKa: 2.20/9.31/NA – pI: 5.76

Legend: pKa (C-ter / N-ter / Side chain) ; MW: monoisotopic molecular weight



pK1 = 2.3
pK2 = 9.6
Glycine
Alanine
Valine
Leucine
Isoleucine
Phenylalanine
Tryptophan
Methionine
Proline

Reference

1- Harpers illustrated Biochemistry

2- Lippincott illustrated reviews integrated systems


Thank
You